



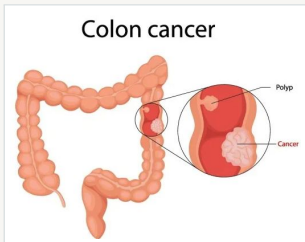
Physics-Informed Neural Network for a Reduced WNT-HOX-Retinoic Acid Model of Colorectal Cancer

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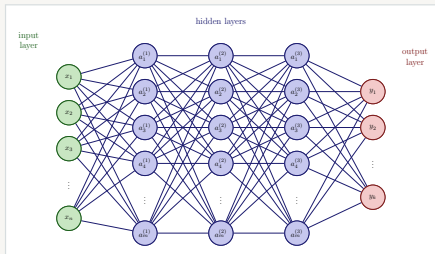
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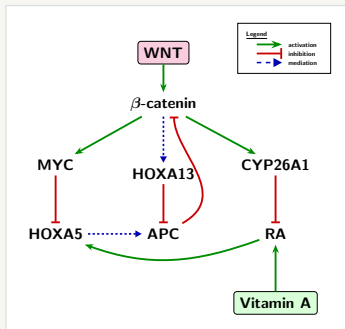
Overview

- Biology/Schematic
- Equations
- Forward PINN
- Inverse PINN
- Inverse Bayesian PINN
- Next Steps

Biological Network Governing WNT–HOX–RA Signaling

Objective

Develop a reduced mathematical model describing the interactions among the WNT, HOX, and Retinoic Acid (RA) pathways during colorectal cancer progression.



State Variables

$$\mathbf{y} = [B(t), P(t), H_5(t), H_{13}(t), M(t), R(t), C(t)]$$

B β -catenin

P APC

H_5 HOXA5

H_{13} HOXA13

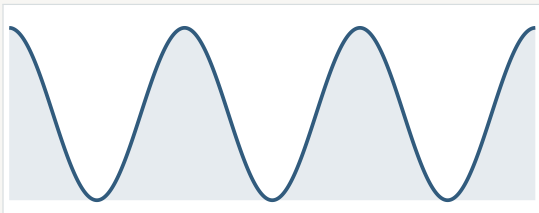
M MYC

R Retinoic Acid

C CYP26A1

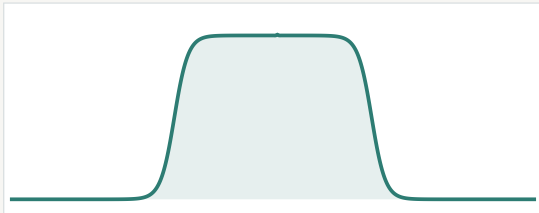
RA Input + Treatment

RA Input



$$A_d \left[1 + \cos\left(\frac{2\pi t}{T_d} - \phi\right) \right]$$

ATRA Treatment



$$u_R(t) = \frac{D}{2} [\tanh q(t - t_1) - \tanh q(t - t_2)]$$

Dimensional Model

$$\frac{dB}{dt} = a_B W + v_{13} \frac{H_{13}^{n_H}}{K_{13}^{n_H} + H_{13}^{n_H}} - d_B B - k_{PB} PB - v_5 \frac{H_5 B}{K_5 + B},$$

$$\frac{dP}{dt} = a_P \frac{1 + \rho_5 H_5}{1 + \rho_B B + \rho_{13} H_{13}} - \left(d_{P0} + d_{P1} (1 - \Theta_P) \right) P,$$

$$\frac{dH_5}{dt} = a_5 + v_R \frac{R}{K_R + R} - d_5 H_5 - v_M \frac{M H_5}{K_M + M},$$

$$\frac{dH_{13}}{dt} = a_{13} + v_{B13} \frac{B^{n_B}}{K_{B13}^{n_B} + B^{n_B}} + v_{M13} \frac{M^{n_M}}{K_{M13}^{n_M} + M^{n_M}} - d_{13} H_{13},$$

$$\frac{dM}{dt} = a_M + v_{BM} \frac{B^{n_B}}{K_{BM}^{n_B} + B^{n_B}} - d_M M,$$

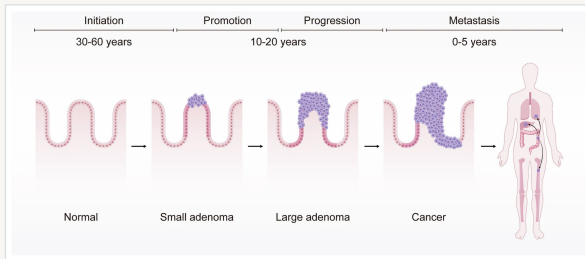
$$\frac{dR}{dt} = u_0 + A_d \left[1 + \cos \left(\frac{2\pi t}{T_d} - \phi \right) \right] + \frac{D}{2} \left[\tanh(q(t - t_1)) - \tanh(q(t - t_2)) \right] - d_R R - k_{CR} CR,$$

$$\frac{dC}{dt} = a_C + v_{RC} \frac{R}{K_{RC} + R} + v_{BC} \frac{B^{n_B}}{K_{BC}^{n_B} + B^{n_B}} - d_C C.$$

4 Regimes

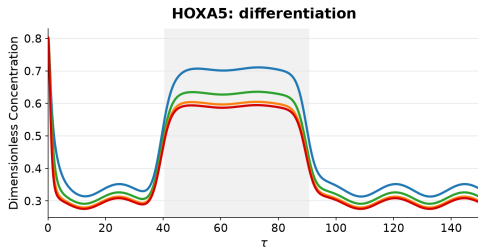
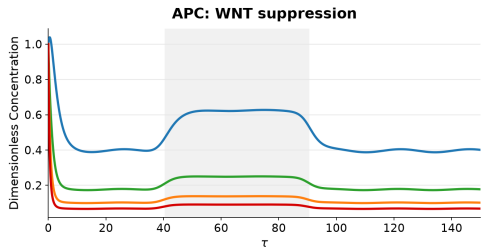
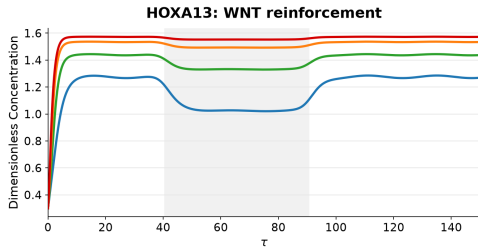
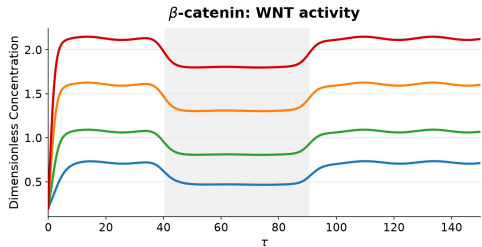
Disease progression is modeled by increasing WNT signaling (W) and progressive loss of APC function (θ_P), mimicking CRC progression.

- “Normal”
 - $W = 0.8, \theta_P = 1.0$
- “Early Adenoma”
 - $W = 1.0, \theta_P = 0.75$
- “Advanced Adenoma”
 - $W = 1.5, \theta_P = 0.5$
- “Severe APC Loss”
 - $W = 2.0, \theta_P = 0.25$

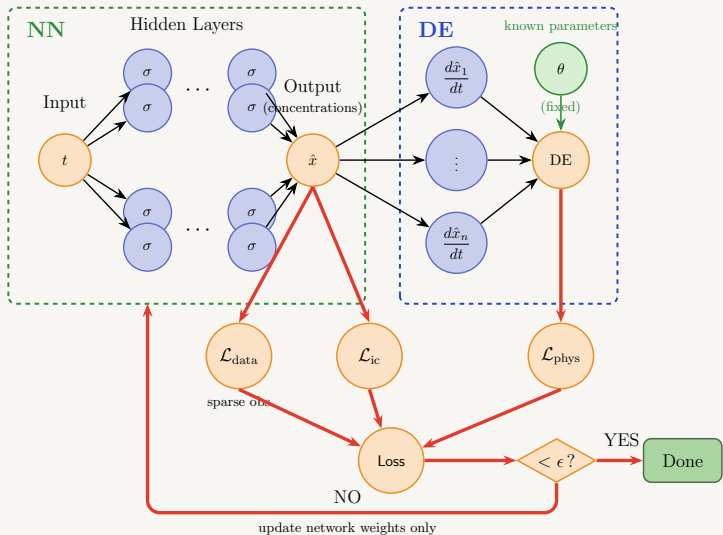


Core WNT-HOX Regulatory Dynamics

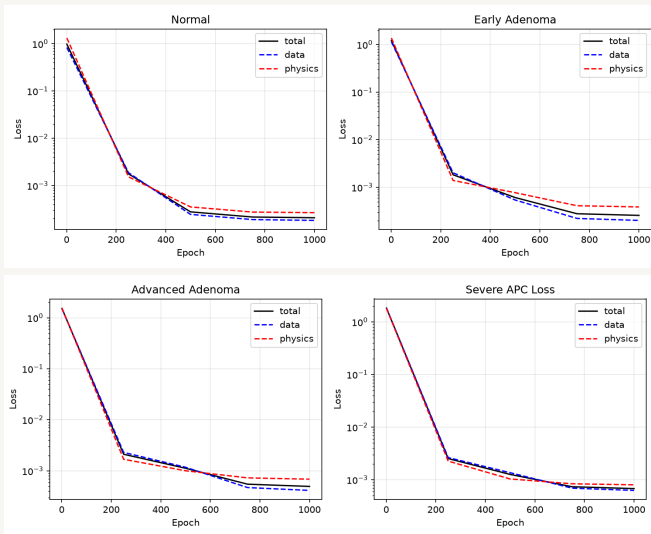
Normal Early Adenoma Advanced Adenoma Severe APC Loss ATRA Window



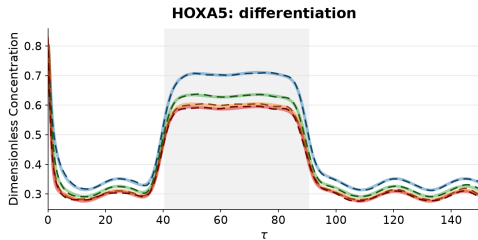
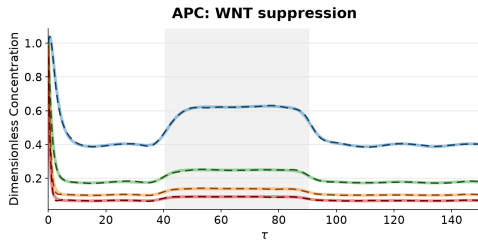
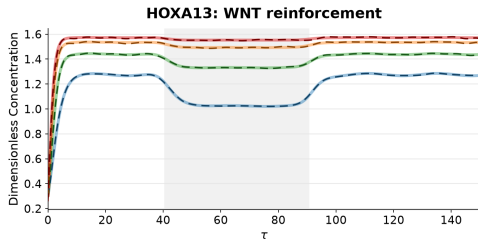
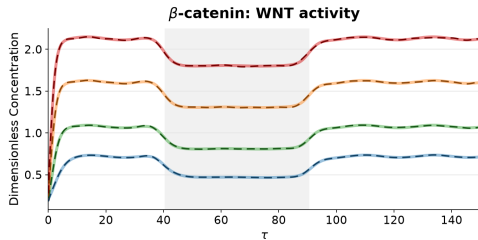
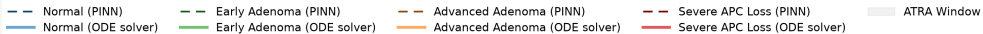
Forward PINN



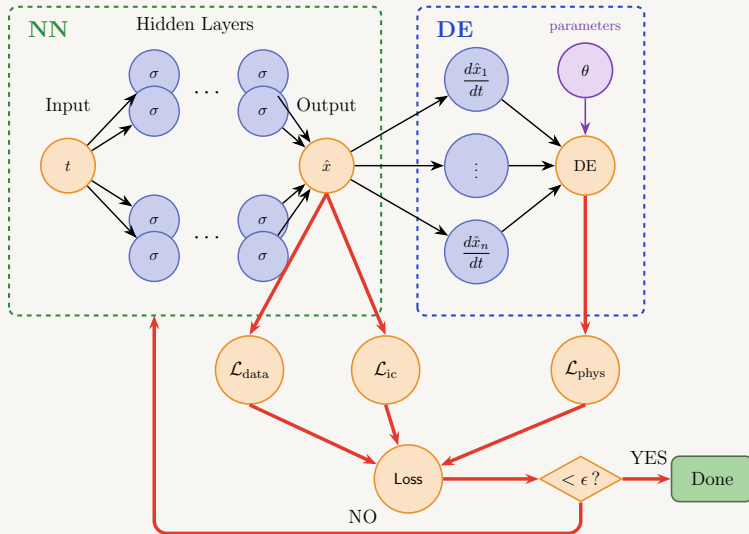
Forward PINN Loss



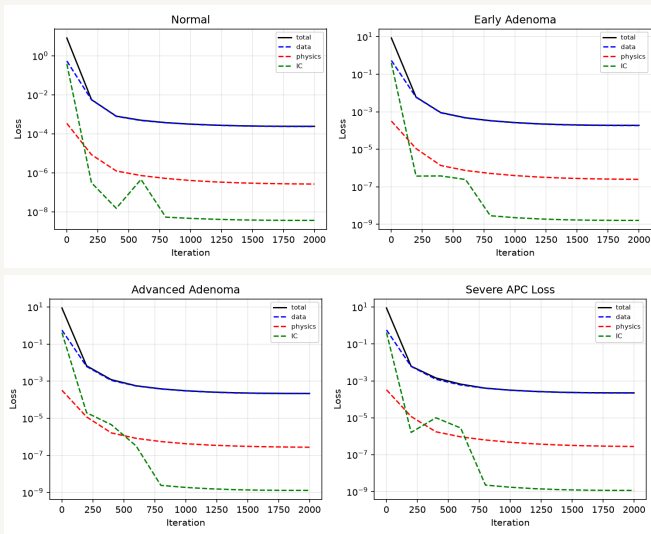
Core WNT-HOX Regulatory Dynamics via Forward PINN



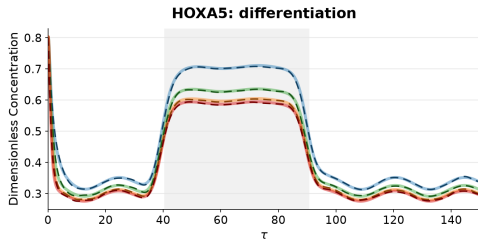
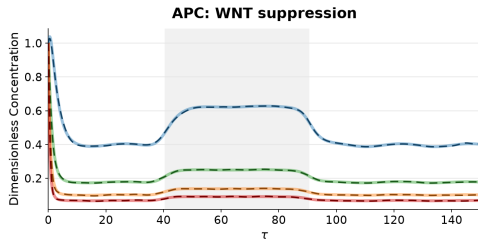
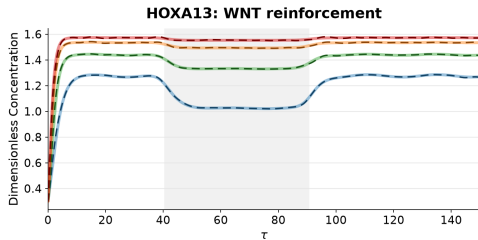
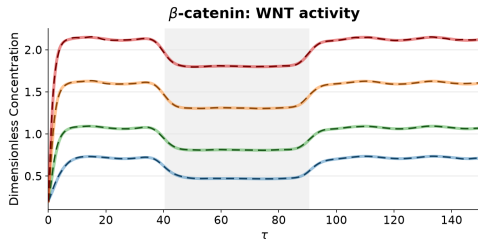
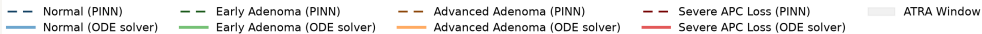
Inverse PINN



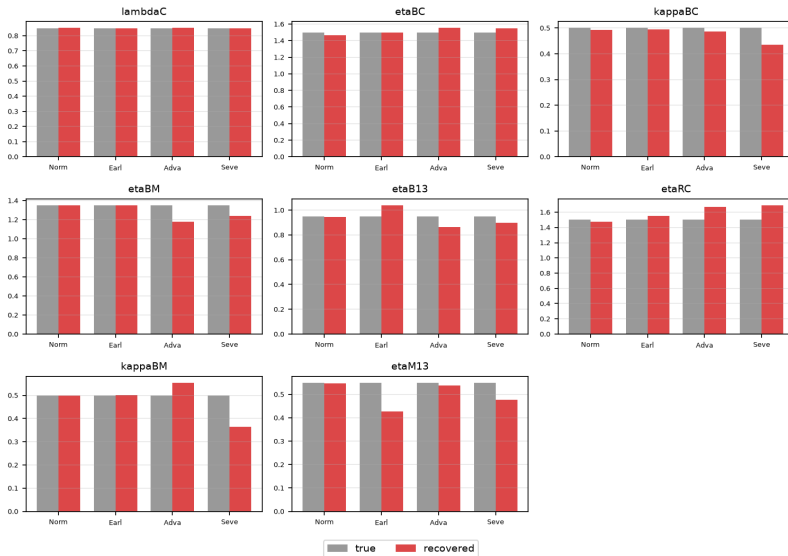
Inverse PINN Loss



Core WNT-HOX Regulatory Dynamics via Inverse PINN

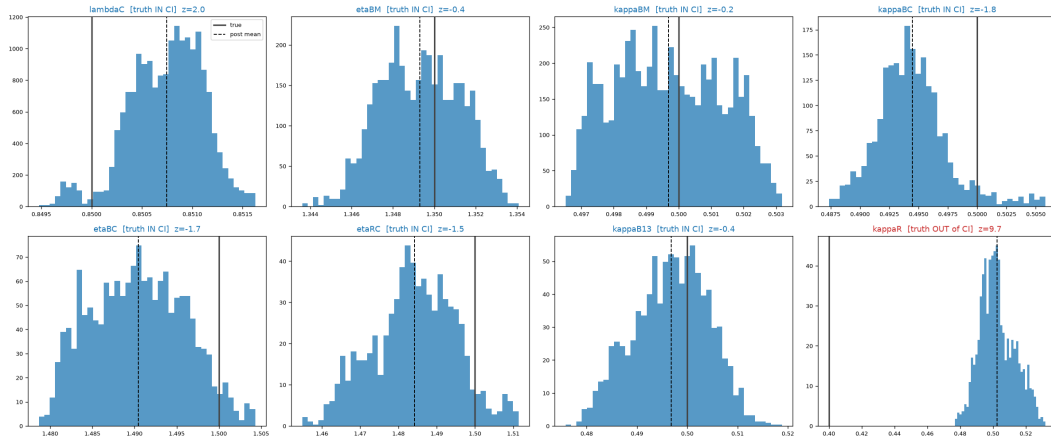


Inverse PINN — true vs recovered (8 best-converging params)

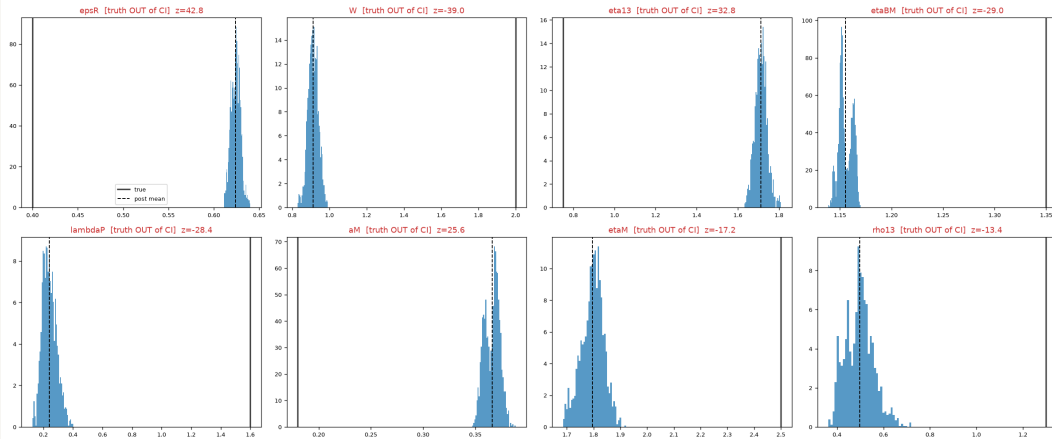


Inverse Bayesian PINN - uncertainty quantification

Normal — posterior marginals (top-8 most sensitive) (blue = truth inside 95% CI, red = confidently wrong) coverage 18/36



Severe APC Loss — posterior marginals (8 worst-recovered) [blue = truth inside 95% CI, red = confidently wrong] coverage 13/36



Parameter Recovery Comparison

	Bayesian inverse PINN	Inverse PINN	ODE-fit
“Normal”	19	17	18/36
“Early Adenoma”	20	16	17/36
“Advanced Adenoma”	11	10	13/36
“Severe APC Loss”	8	7	11/36

Next Steps

- Improve Bayesian PINN
 - Investigate limitations
- Hybrid Mechanistic-Neural Model
- Stochastic Modeling



Key References

1. **Raissi, Perdikaris & Karniadakis (2019)**. Physics-informed neural networks: a deep learning framework for solving forward and inverse problems involving nonlinear PDEs. *J. Comput. Phys.* **378**, 686–707.

Foundational PINN formulation — the data + physics-residual loss used in every model here.

2. **Yazdani, Lu, Raissi & Karniadakis (2020)**. Systems biology informed deep learning for inferring parameters and hidden dynamics. *PLoS Comput. Biol.* **16**(11), e1007575.

Closest template to this work: inverse PINNs on biological ODE networks; the local-sensitivity/Fisher route to identifiability.

3. **Yang, Meng & Karniadakis (2021)**. B-PINNs: Bayesian physics-informed neural networks for forward and inverse PDE problems with noisy data. *J. Comput. Phys.* **425**, 109913.

Basis of the Bayesian inverse PINN: HMC over parameters gives posterior marginals instead of a point estimate.

4. **Raue et al. (2009)**. Structural and practical identifiability analysis of partially observed dynamical models by exploiting the profile likelihood. *Bioinformatics* **25**(15), 1923–1929.

The identifiability verdict — profile likelihood distinguishes the structural from the practical (high-WNT θ_P) wall.

5. **Wang, Yu & Perdikaris (2022)**. When and why PINNs fail to train: a neural tangent kernel perspective. *J. Comput. Phys.* **449**, 110768.

Explains the loss-term imbalance and spectral bias behind the training pathologies; motivates adaptive loss weighting.

6. **Tancik et al. (2020)**. Fourier features let networks learn high frequency functions in low dimensional domains. *NeurIPS* **33**, 7537–7547.

Fourier-feature time embedding — required to capture the circadian RA forcing and the ATRA pulse.

7. **Ji, Qiu, Zhang & Deng (2021)**. Stiff-PINN: physics-informed neural network for stiff chemical kinetics. *J. Phys. Chem. A* **125**(36), 8098–8106.

Diagnoses PINN failure on stiff multiscale kinetics — the same stiffness that shapes the WNT–RA–HOX residual.

Thank you!

Questions? (we will try to answer) 😊

PK calling Amir + Nate to check in on our progress...



4th diet coke of the day + Celsius b/c his
sleep schedule is so cooked

Don't know what in the world is going on
b/c he spent more hours watching the
world cup than working



Thank you very much to the **Provost's Office** for funding
our summer research project!